📃 SLIP 1 – Solutions

✅ Q1: Design a webpage for the following layout For Student profile – where student roll number, name, contact, photo, class and area of interest in column 2. In Colum 1 provide the hyperlinks for Home, Contact us and about us. (HTML)

👉 **Goal:**
Left side = links Right side = student info

💻 Code:

```
<html>
<head>
<title>Student Profile</title>
</head>

<body>

<table border="1" width="100%">
<tr>

<!-- Column 1 -->
<td width="30%">
<a href="#">Home</a><br><br>
<a href="#">Contact Us</a><br><br>
<a href="#">About Us</a>
</td>

<!-- Column 2 -->
<td>
<h2>Student Profile</h2>
Roll No: 101 <br>
Name: User <br>
Contact: 9876543210 <br>
Class: TY Cyber <br>
Area of Interest: Cyber Security <br><br>

<img src="photo.jpg" width="100" height="100">
</td>

</tr>
</table>
```

```
</body>
</html>
```

✅ Q2: Create a login form with a username and password. Display “Welcome” message if username and password is same otherwise display “Invalid username or password” message. (PHP)

👉 **Logic:**
If username == password → Welcome

Else → Invalid 💻 **Code:**

```
<html>
<body>

<form method="post">
Username: <input type="text" name="username"><br><br>
Password: <input type="password" name="password"><br><br>
<input type="submit" value="Login">
</form>
 <?php
if($_POST){
    $u = $_POST['username'];
    $p = $_POST['password'];
     if($u == $p){
echo "Welcome";
    } else {
        echo "Invalid username or password";
    }
}
?>

</body>
</html>
```

📃 SLIP 2 – Solutions

✅ Q1: Write a script to create XML file named “Subject.xml” Stored at least 3 records containing subject id, subject name, class, semester etc.

👉 **Goal:** Create XML with at least 3 records

💻 Code (PHP):

```
<?php


// Create XML object

$xml = new SimpleXMLElement('<Subjects/>');


// Add records

$subject1 = $xml->addChild('Subject');

$subject1->addChild('SubjectID', '101');

$subject1->addChild('SubjectName', 'Web Science');

$subject1->addChild('Class', 'TY');

$subject1->addChild('Semester', '6');


$subject2 = $xml->addChild('Subject');

$subject2->addChild('SubjectID', '102');

$subject2->addChild('SubjectName', 'Cyber Security');

$subject2->addChild('Class', 'TY');

$subject2->addChild('Semester', '6');
```

```
$subject3 = $xml->addChild('Subject');

$subject3->addChild('SubjectID', '103');

$subject3->addChild('SubjectName', 'Data Science');

$subject3->addChild('Class', 'TY');

$subject3->addChild('Semester', '6');


// Save file

$xml->asXML("Subject.xml");


echo "XML file created successfully";


?>
```

⚠ Q2: Using SQL injection attack on Login check whether you can log into another user's account without knowing the correct password.

Objective

Demonstrate how SQL Injection can bypass login authentication without knowing the correct password.

Background: How Login Works Normally

A typical login query looks like this:

`SELECT * FROM users WHERE username = 'test1' AND password = 'mypassword';` If both match → login success. If not → access denied.

Setting Up the Test

Target site: `http://testasp.vulnweb.com/Default.asp` *(intentionally vulnerable for testing)*

Step 1: Create two accounts:

Account	Password
test1	pass123
test2	pass456

Step 2: Try logging into `test1` with a **wrong password** → Access Denied ✅ (expected)

The Attack

Now in the **password field**, instead of the real password, enter:

```
' OR 1=1 --
```

Full input:

- Username: `test1`
- Password: `' OR 1=1 --`

Why This Works

The backend query becomes:

```
SELECT * FROM users
WHERE username = 'test1' AND password = '' OR 1=1 --'
```

Breaking it down:

Part	Effect
`'`	Closes the password string prematurely
`OR 1=1`	Always evaluates to **TRUE**
`--`	Comments out the rest of the query (ignores original closing quote)

So the entire WHERE condition becomes **TRUE**, bypassing the password check entirely → **Login succeeds without the correct password.**

Result

✅ You gain access to `test1`'s account without knowing their password — purely by manipulating the SQL query logic.

Why This is Dangerous (Real World)

- Attacker can log in as **any user**, including admin accounts
- No brute force needed — one crafted input is enough
- Can lead to **full database compromise** if combined with UNION-based injection

---

Mitigation (How to Fix It)

Fix	How
Prepared Statements	`SELECT * FROM users WHERE username = ? AND password = ?`
Parameterized Queries	Separates data from SQL logic
Input Validation	Reject special characters like `'`, `--`, `;`
WAF	Web Application Firewall to detect injection patterns

The root cause is **unsanitized user input being directly concatenated into SQL queries**. Prepared statements completely eliminate this vulnerability.

📜 SLIP 6

✅ Q1: Write a PHP program to write a program for find the given number is palindrome or not. (PHP)

👉 **Goal:**
Check number = same forward & backward

💻 Code:

```
<?php

$num = 121;
$rev = strrev($num);
 if($num == $rev){
echo "Palindrome";
} else {
    echo "Not Palindrome";
}

?>
```

✅ Q2: Create a XML document which contains details of cars car like: id, company name, model engine and mileage and display the same as a table by using element by using XSLT.

👉 **Goal:**

- Create XML
- Display as table using XSLT

💻 XML (cars.xml):

```
<?xml version="1.0"?>
<Cars>

<Car>
```

```
    <ID>1</ID>
    <Company>Toyota</Company>
    <Model>Innova</Model>
    <Engine>2000cc</Engine>
    <Mileage>15kmpl</Mileage>
</Car>

<Car>
    <ID>2</ID>
    <Company>Honda</Company>
    <Model>City</Model>
    <Engine>1500cc</Engine>
    <Mileage>18kmpl</Mileage>
</Car>

</Cars>
```

💻 XSLT (cars.xsl):

```
<?xml version="1.0"?> <xsl:stylesheet
version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:template match="/">

<html>
<body>
<table border="1">
<tr>
<th>ID</th><th>Company</th><th>Model</th>
<th>Engine</th><th>Mileage</th>
</tr>

<xsl:for-each select="Cars/Car">
<tr>
<td><xsl:value-of select="ID"/></td>
<td><xsl:value-of select="Company"/></td>
<td><xsl:value-of select="Model"/></td>
<td><xsl:value-of select="Engine"/></td>
<td><xsl:value-of select="Mileage"/></td>
</tr>
</xsl:for-each>

</table>
</body>
</html>

</xsl:template>
</xsl:stylesheet>
```

📃 SLIP 8

✅ Q1: Create an html page with 7 separate lines in different colors. State color of each line in its text.

👉 **Goal:**
7 lines, each with different color + mention color name

💻 Code:

```
<html>
<body>

<p style="color:red;">This is Red</p>
<p style="color:blue;">This is Blue</p>
<p style="color:green;">This is Green</p>
<p style="color:yellow;">This is Yellow</p>
<p style="color:orange;">This is Orange</p>
<p style="color:purple;">This is Purple</p>
<p style="color:brown;">This is Brown</p>

</body>
</html>
```

✅ Q2: Design an application to accept username and password. While accepting password check whether is it made of numbers, alphabets, special symbols and password length should be greater than 8. If password is as per requirement, then print “You enter red strong password” else print error message “Enter complex password”.

☞ Same logic as Slip 4)

💻 Code:

```
<html>
<body>

<h2>Password Validation</h2>

<form method="post"> Enter
Password:
<input type="password" name="password"><br><br>

<input type="submit" value="Check Password">
</form>

<?php if($_POST){

   $password = $_POST['password'];

   if(strlen($password)  >  8  &&
preg_match("/[A-Za-z]/",  $password)  &&
preg_match("/[0-9]/", $password) &&
      preg_match("/[@$!%*#?&]/", $password)){

       echo "You entered strong password";
   } else {
       echo "Enter complex password";
   }
}
?>

</body>
</html>
```

📜 SLIP 9

✅ Q1:

Design a HTML form for Bus Ticket Reservation consisting of fields for Name, Address, contact no, source station (Dropdown list), Destination station, Date of booking, date of journey, no of passenger, name of passenger, gender of passenger etc. (Use proper alignment)

💻 Answer:

```
<html>
<body>

<h2>Bus Ticket Reservation</h2>

<form>

Name: <input type="text"><br><br>
Address: <input type="text"><br><br>
Contact No: <input type="text"><br><br>
Source:
<select>
<option>Pune</option>
<option>Mumbai</option>
<option>Nashik</option>
</select><br><br>

Destination:
<select>
<option>Pune</option>
<option>Mumbai</option>
<option>Nashik</option>
</select><br><br>

Date of Booking: <input type="date"><br><br>
Date of Journey: <input type="date"><br><br>

No. of Passengers: <input type="number"><br><br>

Passenger Name: <input type="text"><br><br>
Gender:
<input type="radio" name="g">Male
<input type="radio" name="g">Female<br><br>

<input type="submit" value="Book Ticket">

</form>

</body>
</html>
```

✅ Q2:

Design an HTML form for customer registration visiting a departmental store. Form should consist of fields such as name, contact no, gender, preferred days of purchasing, favorite item (to be selected from a list of items), suggestions etc. Provide submit and reset buttons.

💻 Answer:

```
<html>
<body>

<h2>Customer Registration</h2>

<form>

Name: <input type="text"><br><br>
Contact No: <input type="text"><br><br>
Gender:
<input type="radio" name="g">Male
<input type="radio" name="g">Female<br><br>

Preferred Days:
<input type="checkbox">Monday
<input type="checkbox">Weekend<br><br>

Favorite Item:
<select>
<option>Clothes</option>
<option>Electronics</option>
<option>Groceries</option>
</select><br><br>

Suggestions:<br>
<textarea></textarea><br><br>

<input type="submit" value="Submit">
<input type="reset" value="Reset">

</form>

</body>
</html>
```

📜 SLIP 10

✅ Q1:

Write a script that generates an XML file named “Student.xml” containing at least 5 student records. Each record should include: Student ID, Student Name, Age, Gender, Program

---

💻 Answer (XML):

```
<?php

$xml = new SimpleXMLElement('<Students/>');

// Student 1
$s1 = $xml->addChild('Student');
$s1->addChild('ID', '1');
$s1->addChild('Name', 'User');
$s1->addChild('Age', '20');
$s1->addChild('Gender', 'Male');
$s1->addChild('Program', 'Cyber');

// Student 2
$s2 = $xml->addChild('Student');
$s2->addChild('ID', '2');
$s2->addChild('Name', 'Rahul');
$s2->addChild('Age', '21');
$s2->addChild('Gender', 'Male');
$s2->addChild('Program', 'IT');

// Student 3
$s3 = $xml->addChild('Student');
$s3->addChild('ID', '3');
$s3->addChild('Name', `User1');
$s3->addChild('Age', '20');
$s3->addChild('Gender', 'Female');
$s3->addChild('Program', 'CS');

// Student 4
$s4 = $xml->addChild('Student');
$s4->addChild('ID', '4');
$s4->addChild('Name', 'Anjali');
$s4->addChild('Age', '22');
$s4->addChild('Gender', 'Female');
$s4->addChild('Program', 'AI');

// Student 5
$s5 = $xml->addChild('Student');
$s5->addChild('ID', '5');
$s5->addChild('Name', 'Amit');
$s5->addChild('Age', '21');
$s5->addChild('Gender', 'Male');
$s5->addChild('Program', 'Data Science');
```

```
// Save file
$xml->asXML("Student.xml");
 echo "XML
Created";

?>
```

✅ Q2:

Write a script to create an XML file called “Employees.xml” containing records for 5 employees. Each record should contain: Employee ID, Name, Position, Department, Date of Joining

---

💻 Answer (PHP):

```
<?php

$xml = new SimpleXMLElement('<Employees/>');

// Employee 1
$e1 = $xml->addChild('Employee');
$e1->addChild('ID', '101');
$e1->addChild('Name', 'User2');
$e1->addChild('Position', 'Developer');
$e1->addChild('Department', 'IT');
$e1->addChild('DOJ', '2023-01-10');

// Employee 2
$e2 = $xml->addChild('Employee');
$e2->addChild('ID', '102');
$e2->addChild('Name', 'Rahul');
$e2->addChild('Position', 'Manager');
$e2->addChild('Department', 'HR');
$e2->addChild('DOJ', '2022-05-15');

// Employee 3
$e3 = $xml->addChild('Employee');
$e3->addChild('ID', '103');
$e3->addChild('Name', 'User5
4');
$e3->addChild('Position', 'Analyst');
$e3->addChild('Department', 'Finance');
$e3->addChild('DOJ', '2023-03-20');
```

```
// Employee 4
$e4 = $xml->addChild('Employee');
$e4->addChild('ID', '104');
$e4->addChild('Name', 'Anjali');
$e4->addChild('Position', 'Tester');
$e4->addChild('Department', 'QA');
$e4->addChild('DOJ', '2024-02-01');

// Employee 5
$e5 = $xml->addChild('Employee');
$e5->addChild('ID', '105');
$e5->addChild('Name', 'Amit');
$e5->addChild('Position', 'Support');
$e5->addChild('Department', 'IT');
$e5->addChild('DOJ', '2023-07-12');

// Save file
$xml->asXML("Employees.xml");

echo "XML Created";

?>
```

📜 SLIP 11

✅ Q1:

Write a PHP script to accept username and password. If in the first three chances, username and password entered is correct then display second form with “Welcome message” otherwise display error message. [Use Session]

💻 Answer (PHP with Session):

```php
<?php
session_start();

// correct credentials
$correct_user = "admin";
$correct_pass = "1234";

// attempt count
if(!isset($_SESSION['attempt'])){
    $_SESSION['attempt'] = 0;
}
```

```
if($_POST){
    $u = $_POST['username'];
    $p = $_POST['password'];

    if($u == $correct_user && $p == $correct_pass){
echo "<h2>Welcome</h2>";         session_destroy();
    } else {
        $_SESSION['attempt']++;

        if($_SESSION['attempt'] < 3){
            echo "Invalid Login. Attempts: " . $_SESSION['attempt'];
        } else {
            echo "Account Blocked";
        }
    }
}
?>

<form method="post">
Username: <input type="text" name="username"><br><br>
Password: <input type="password" name="password"><br><br>
<input type="submit" value="Login">
</form>
```

✅ Q2:

Write a PHP script to accept Employee details (Eno, Ename, Address) on first page. On second page accept earning (Basic, DA, HRA). On third page print Employee information (Eno, Ename, Address, Basic, DA, HRA, Total)

☞ 3 Pages concept (keep it simple)

💻 Page 1 (employee.php)

```
<form action="page2.php" method="post">
Eno: <input type="text" name="eno"><br>
Name: <input type="text" name="name"><br> Address:
<input type="text" name="address"><br>
<input type="submit" value="Next">
</form>
```

💻 Page 2 (page2.php)

```
<form action="page3.php" method="post">
<input type="hidden" name="eno" value="<?php echo $_POST['eno']; ?>">
```

```
<input type="hidden" name="name" value="<?php echo $_POST['name']; ?>">
<input type="hidden" name="address" value="<?php echo $_POST['address'];
?>">

Basic: <input type="text" name="basic"><br>
DA: <input type="text" name="da"><br>
HRA: <input type="text" name="hra"><br>

<input type="submit" value="Submit">
</form>
```

💻 Page 3 (page3.php)

```
<?php

$eno = $_POST['eno'];
$name = $_POST['name'];
$address = $_POST['address'];

$basic = $_POST['basic'];
$da = $_POST['da'];
$hra = $_POST['hra'];

$total = $basic + $da + $hra;
 echo "Eno: $eno <br>"; echo
"Name: $name <br>"; echo
"Address: $address <br>"; echo
"Total Salary: $total"; ?>
```

📜 SLIP 12 – Complete (Questions + Answers)

✅ Q1:

Design HTML form to accept dimension of a cylinder and write a PHP script to calculate the area and volume of Cylinder using a function.

💻 Answer:

🔹 *HTML + PHP:*

```
<html>
<body>
```

```
<form method="post">
Radius: <input type="text" name="r"><br><br>
Height: <input type="text" name="h"><br><br>
<input type="submit" value="Calculate">
</form>
 <?php
function cylinder($r, $h){
    $area = 2 * 3.14 * $r * ($r + $h);
    $volume = 3.14 * $r * $r * $h;
     echo "Surface Area: $area
<br>";     echo "Volume: $volume";
}
if($_POST){
    $r = $_POST['r'];
$h = $_POST['h'];
cylinder($r, $h);
}
?>

</body>
</html>
.
```

✅ Q2:

Design HTML form to accept two strings and following three options (use radio button) and write a PHP script to perform: a) Compare 2 strings
b) Convert string into Uppercase
c) Convert string into Lowercase

💻 Answer:

```
<html>
<body>

<form method="post">
String 1: <input type="text" name="s1"><br><br>
String 2: <input type="text" name="s2"><br><br>

<input type="radio" name="op" value="compare">Compare<br>
<input type="radio" name="op" value="upper">Uppercase<br>
<input type="radio" name="op" value="lower">Lowercase<br><br>

<input type="submit" value="Submit">
</form>
 <?php
if($_POST){
    $s1 = $_POST['s1'];
    $s2 = $_POST['s2'];
    $op = $_POST['op'];
```

```
     if($op ==
"compare"){
if($s1 == $s2){
            echo "Strings are Equal";
        } else {
            echo "Strings are Not Equal";
        }
    }
    if($op == "upper"){
echo strtoupper($s1);
    }
    if($op == "lower"){
echo strtolower($s1);
    }
}
?>

</body>
</html>
```

📜 SLIP 13 – Complete (Questions + Answers)

✅ Q1:

Design a HTML form to accept a number and write a PHP script to display each digit of a number in words.

💻 Answer:

```
<html>
<body>

<form method="post">
Enter Number: <input type="text" name="num"><br><br>
<input type="submit" value="Convert">
</form>
 <?php
if($_POST){
    $num = $_POST['num'];

    $words = [
```

```
        '0'=>"Zero", '1'=>"One", '2'=>"Two", '3'=>"Three",
        '4'=>"Four", '5'=>"Five", '6'=>"Six",
        '7'=>"Seven", '8'=>"Eight", '9'=>"Nine"
    ];

    $digits = str_split($num);
     foreach($digits as $d){
echo $words[$d] . " ";
    }
}
?>

</body>
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
</html>
```

✅ Q2:

Design a HTML form to accept a number and write a PHP script to display each digit of a number in words.

☞ ⚠ Same question repeated → just rewrite Q1 answer

📜 SLIP 14

✅ Q1:

Create a PHP program where the user can select items from a product list (e.g., books, pens, or bags) and input quantities. Display:

- Selected items with unit price
- Total quantity
- Total cost

💻 Answer:

```
<html>
<body>

<form method="post">

Book (100): <input type="number" name="book"><br><br>
Pen (20): <input type="number" name="pen"><br><br>
Bag (500): <input type="number" name="bag"><br><br>
```

```
<input type="submit" value="Calculate">

</form>
 <?php
if($_POST){

    $book = $_POST['book'];
    $pen = $_POST['pen'];
    $bag = $_POST['bag'];

    $total_qty = $book + $pen + $bag;
    $total_cost = ($book*100) + ($pen*20) + ($bag*500);

    echo "Total Quantity: $total_qty <br>";
echo "Total Cost: $total_cost";

}
?>

</body>
</html>
```

✅ Q2:

Explain OWASP Top 10 threats for web (introduction)

Explain in detail in exam if needed

✍ Answer:

OWASP Top 10 is a list of most critical web security risks.

Some important threats:

1. **Injection (SQL Injection)**
   ☞ Attacker inserts malicious SQL queries
2. **Broken Authentication**
   ☞ Weak login system allows unauthorized access
3. **Sensitive Data Exposure**
   ☞ Data like passwords not properly protected
4. **Broken Access Control**
   ☞ Users access unauthorized resources
5. **Security Misconfiguration**
   ☞ Improper server settings
6. **Cross-Site Scripting (XSS)**

☞ Injecting malicious scripts into web pages

7. **Insecure Deserialization**
   ☞ Unsafe data processing
8. **Using Components with Vulnerabilities**
   ☞ Outdated libraries
9. **Insufficient Logging & Monitoring**
   ☞ Attacks not detected
10. **Server-Side Request Forgery (SSRF)**
    ☞ Server forced to make malicious requests

SLIP 15

✓ Q1:

Write a short report explaining: a)
The identified threat
b) How the threat was exploited
c) The steps you took to mitigate the vulnerability

✍ Answer:

a) Identified Threat:
The identified threat is **SQL Injection**, where an attacker inserts malicious SQL queries into input fields.

---

b) Exploitation:

The attacker enters input like `' OR 1=1 --` in the login form. This bypasses authentication and allows unauthorized access.

---

c) Mitigation Steps:

- Use prepared statements
- Validate user input
- Use parameterized queries

- Apply proper authentication
- Use web application firewall

✅ Q2:

Write a script to create an XML file called “Timetable.xml” containing 3 records for a weekly class timetable. Each record should include:

- Day
- Class Name
- Instructor
- Time Slot
- Classroom**

💻 Answer (XML):

```
<?php

$xml = new SimpleXMLElement('<Timetable/>');


// Record 1

$c1 = $xml->addChild('Class');

$c1->addChild('Day', 'Monday');

$c1->addChild('ClassName', 'Web Science');

$c1->addChild('Instructor', 'Prof. A');

$c1->addChild('Time', '10:00 AM');

$c1->addChild('Room', '101');


// Record 2

$c2 = $xml->addChild('Class');
```

```
$c2->addChild('Day', 'Tuesday');
$c2->addChild('ClassName', 'Cyber Security');
$c2->addChild('Instructor', 'Prof. B');
$c2->addChild('Time', '12:00 PM');
$c2->addChild('Room', '102');

// Record 3
$c3 = $xml->addChild('Class');
$c3->addChild('Day', 'Wednesday');
$c3->addChild('ClassName', 'Data Science');
$c3->addChild('Instructor', 'Prof. C');
$c3->addChild('Time', '2:00 PM'); $c3->addChild('Room', '103');

// Save file
$xml->asXML("Timetable.xml");

echo "XML Created";

?>
```

📜 SLIP 16

✅ Q1:

Using SQL injection attack on Login checks whether you can log into another user's account without knowing the correct password.

💻 Answer:

Same as above question in the slip 2 q2

✅ Q2:

Write a short note on Burp Suite Scanner with an example.

✍ Answer:

Burp Suite Scanner is a tool used to find security vulnerabilities in web applications.
It automatically scans websites for issues like SQL Injection, XSS, etc.

🔹 Features:

- Automated scanning
- Detects vulnerabilities
- Intercepts HTTP requests
- Provides detailed reports

🔹 Example:

While testing a login page, Burp Suite Scanner detects a **SQL Injection vulnerability** in the username field.

📝 Exam line:

☞ Burp Suite Scanner is used to identify web vulnerabilities automatically.

📃 SLIP 17 – Complete (Questions + Answers)

✅ Q1:

Create a PHP program where the user can select items from a product list (e.g., books, pens, or bags) and input quantities. Display:

- Selected items with their unit price
- Total quantity
- Total cost**

☞ ⚠ Same as Slip 14

💻 Answer:

```
<html>
<body>

<form method="post">

Book (100): <input type="number" name="book"><br><br>
Pen (20): <input type="number" name="pen"><br><br>
Bag (500): <input type="number" name="bag"><br><br>

<input type="submit" value="Calculate">

</form>
 <?php
if($_POST){

    $book = $_POST['book'];
    $pen = $_POST['pen'];
    $bag = $_POST['bag'];

    $total_qty = $book + $pen + $bag;
    $total_cost = ($book*100) + ($pen*20) + ($bag*500);
     echo "Total Quantity: $total_qty
<br>";     echo "Total Cost: $total_cost";
}
?>

</body>
```

```
</html>
```

✅ Q2:

Write a script to generate an XML file called "CourseRegistrations.xml" that stores 5 records of students registering for courses. Each record should contain:

- Registration ID □ Student Name □ Course Name
- Instructor
- Date of Registration**

💻 Answer (PHP):

```
<?php

$xml = new SimpleXMLElement('<CourseRegistrations/>');

// Record 1
$r1 = $xml->addChild('Registration');
$r1->addChild('ID', '1');
$r1->addChild('StudentName', 'User3');
$r1->addChild('CourseName', 'Cyber Security');
$r1->addChild('Instructor', 'Prof. A');
$r1->addChild('Date', '2024-01-10');

// Record 2
$r2 = $xml->addChild('Registration');
$r2->addChild('ID', '2');
$r2->addChild('StudentName', 'Rahul');
```

```
$r2->addChild('CourseName', 'Web Science');
$r2->addChild('Instructor', 'Prof. B');
$r2->addChild('Date', '2024-01-12');

// Record 3
$r3 = $xml->addChild('Registration');
$r3->addChild('ID', '3');
$r3->addChild('StudentName', 'User’);
$r3->addChild('CourseName', 'Data Science');
$r3->addChild('Instructor', 'Prof. C');
$r3->addChild('Date', '2024-01-15');

// Record 4
$r4 = $xml->addChild('Registration');
$r4->addChild('ID', '4');
$r4->addChild('StudentName', 'Anjali');
$r4->addChild('CourseName', 'AI');
$r4->addChild('Instructor', 'Prof. D');
$r4->addChild('Date', '2024-01-18');

// Record 5
$r5 = $xml->addChild('Registration');
$r5->addChild('ID', '5');
$r5->addChild('StudentName', 'Amit');
```

$r5->addChild('CourseName', 'Cloud Computing');

$r5->addChild('Instructor', 'Prof. E');

$r5->addChild('Date', '2024-01-20');

// Save file

$xml->asXML("CourseRegistrations.xml");

echo "XML Created";

?>

📃 SLIP 21

✅ Q1:

Design HTML form to accept a number and write a PHP script to calculate the factorial of a number using function.

💻 Answer:

```
<html>
<body>

<form method="post">
Enter Number: <input type="text" name="num"><br><br>
<input type="submit" value="Calculate">
</form>
 <?php
function factorial($n){
$fact = 1;
    for($i=1; $i<=$n; $i++){
        $fact *= $i;
    }
    return $fact;
}
if($_POST){
```

```
    $num = $_POST['num'];
    echo "Factorial = " . factorial($num);
}
?>

</body>
</html>
```

✅ Q2:

Write a short note on OWASP ZAP scanner and give one example.

✍ Answer:

OWASP ZAP (Zed Attack Proxy) is a web security testing tool used to find vulnerabilities in web applications.

🔹 Features:

- Intercepts HTTP requests
- Automated vulnerability scanning
- Detects SQL Injection, XSS
- Free and open-source

🔹 Example:

While testing a website, ZAP detects a **Cross-Site Scripting (XSS)** vulnerability in a search input field.

📄 SLIP 22

✅ Q1:

Write down OWASP Top 10 threats for API.

✍ Answer (Short + Exam Ready) (if asked for detailed answer write in detail):

OWASP API Top 10 lists common API security risks:

1. **Broken Object Level Authorization (BOLA)**
   ☞ Accessing other user's data
2. **Broken Authentication**
   ☞ Weak login system
3. **Broken Object Property Level Authorization**
   ☞ Unauthorized data modification
4. **Unrestricted Resource Consumption**
   ☞ API abuse (DoS)
5. **Broken Function Level Authorization**
   ☞ Accessing restricted functions
6. **Mass Assignment**
   ☞ Changing protected fields
7. **Security Misconfiguration**
   ☞ Improper API setup
8. **Injection (SQL, NoSQL)**
   ☞ Malicious queries
9. **Improper Asset Management**
   ☞ Old/unused APIs exposed
10. **Insufficient Logging & Monitoring**
    ☞ Attacks not detected

✅ Q2:

Design an application to accept username and password. While accepting password check whether it contains numbers, alphabets, special symbols and length > 8. If valid → "strong password", else → "Enter complex password".

💻 Answer:

```
<html>
<body>

<form method="post">
Username: <input type="text" name="user"><br><br>
Password: <input type="password" name="pass"><br><br>
<input type="submit" value="Check">
</form>
 <?php
if($_POST){

    $pass = $_POST['pass'];

    if(strlen($pass) > 8 &&        preg_match("/[A-
Za-z]/", $pass) &&        preg_match("/[0-9]/",
$pass) &&        preg_match("/[@$!%*#?&]/", $pass)){

        echo "You entered strong password";
    } else {
        echo "Enter complex password";
    }
}
?>

</body>
</html>
```